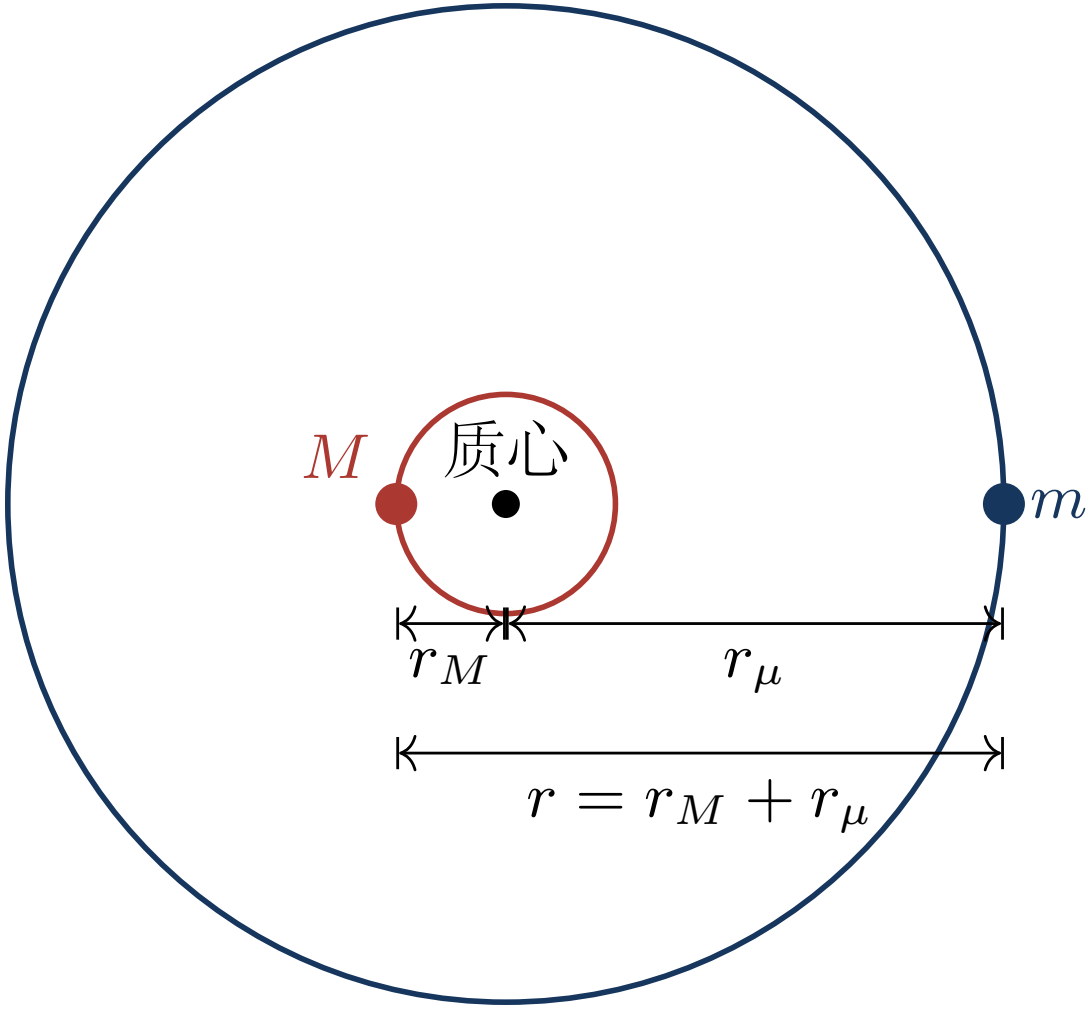

$$Mr_M = mr_\mu$$

$$\mu_r = mM/(m+M)$$